

Learning-Based Predictive Impedance Control Towards Safe Predefined-Time Physical Robotic Interaction

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Abstract—Impedance control can be achieved within a model predictive control (MPC) framework for optimization and constraint compliance. However, user-defined or optimization-derived impedance models can be too conservative to achieve a timely convergence, or too aggressive to ensure safety. To address this, an MPC-based impedance control framework with learning-based tuning for predefined-time (PdT) convergence is proposed. On the low level, the framework dynamically selects between a task-oriented and a safety-oriented impedance model based on real-time interaction force modeling and safety assessments, ensuring optimal performance and maintaining safety while interacting with unknown and complex environments. On the high level, the framework achieves PdT convergence via reinforcement learning for meta-parameter tuning, allowing users to specify the desired convergence time upper bound. Lastly, the superiority of the proposed framework is validated on interaction safety and PdT convergence via experiments.

I. INTRODUCTION

Over the past few decades, robotics has evolved with a growing emphasis on enabling robots to perform tasks in complex and unknown environments. This progression has been driven by the increasing need for human-robot collaboration, where robots work alongside humans in various scenarios, e.g., collaborative assembly [1] and medical operations [2]. Yet, designing control systems that ensure a safe and effective physical robotic interaction is still challenging.

One fundamental control for physical interaction is impedance control (IC) [3]. IC indirectly regulates the interaction force and robot position based on a virtual impedance model, enabling the robot to comply with the interaction environment and avoid potential risks. Upon IC, Variable Impedance Control (VIC) is proposed for a variable impedance model [4]. Many methods achieve VIC, including learning from demonstration [5], [6], reinforcement learning (RL) [7], [8], and linear quadratic regulators [9], [10].

Passivity is a crucial property for VIC. A passive system cannot generate energy internally [11], preventing uncontrolled growth of forces or motions, and thus ensuring stability and enhancing safety. Upon the classical passive control, energy-aware control can be employed to enhance control flexibility, such that the passivity is maintained by an augmented system [12], [13].

Besides the impedance variation from VIC, constraints are desired for a safe robotic interaction [14]. For this purpose,

Model Predictive Control (MPC) is usually adopted. In [15], IC is realized within an MPC framework. In [16], MPC incorporates VIC for minimizing the tracking error. VIC is learned and combined with MPC in [17]. In [18], MPC optimizes the robot trajectory and impedance model. In [19], MPC is combined with VIC to regulate the robot behavior to enhance interaction safety.

Some tasks require an explicit convergence time, while the convergence of impedance models is only implicitly influenced by the impedance parameter and interaction environment. Predefined-time (PdT) convergence is a more controllable property for designing convergence behavior. Specifically, PdT convergence guarantees that the system will reach a steady state (or its neighborhood) within a user-defined time [20], [21]. Several recent works have designed PdT controllers for real-world applications, e.g., manipulators [22] and a fire-fighting robot [23].

Practically achieving the PdT convergence relies on the appropriate tuning of a key meta-parameter in this work. Many works tune controllers by RL, e.g., [24] for PID, [25] for MPC, and [26] for sliding mode control. It will be demonstrated later that the tuning problem of this work can be fitted into a contextual bandit RL problem [27], where the agent tries to collect the maximum reward by selecting the best action (arm) from a finite set considering the current context. To balance the exploration and exploitation, the Upper Confidence Bound (UCB) algorithm is adopted in bandit problems [28], [29]. A neural network (NN) is adopted in [30] and [31] for nonlinear and complex relationships between context and reward.

To the best of our knowledge, this is the first work to propose an optimization-based control framework that safely achieves PdT convergence via a novel MPC-based impedance control with learning-based meta-parameter tuning.

The main contributions of this work are:

- 1) proposing an MPC-based safe and passive IC framework for physical robotic interactions;
- 2) achieving a practical PdT convergence based on the computed control inputs from the MPC and the proposed time scaling;
- 3) learning a tuning schedule on the controller meta-parameter to balance between the interaction safety and the performance of PdT convergence.

The rest of this work is structured as follows. Section II formulates the fundamentals for the proposed control framework developments in Section III. Section IV presents the experiments to validate the superiority of the proposed framework. Finally, Section V concludes the whole work.

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B. Safe Impedance Variation and Passivity

In Subsection III-A, the MPC-based framework optimizes based on fixed impedance parameters m , d , and k . Impedance variation and passivity are desired for interaction safety. Thus, define a second impedance model:

$$m\ddot{e} + d_+\dot{e} = -e_f, \quad (10)$$

where $d_+ > d$ is a safer damping coefficient. This model is a pure force control model with stiffness equal to zero to overcome disturbances.

The impedance model varies between (4) and (10) based on the real-time safety situation. The ISO standard [34] states that safety in physical robotic interactions is related to (i) interaction force, (ii) velocity, and (iii) energy flow. Following this standard, a safety indicator is defined as:

$$\begin{aligned} s &= s(e_f, \dot{e}, p_w) = s_f s_v s_p, \\ s_f &= \sigma(e_f, -w_f, 0)[1 - \sigma(e_f, f_r - w_f, f_r)], \\ s_v &= \sigma(\dot{e}, -\dot{e}_b, -\dot{e}_b + w_v)[1 - \sigma(\dot{e}, \dot{e}_b - w_v, \dot{e}_b)], \\ s_p &= \sigma(p_w, -p_b, -p_b + w_p)[1 - \sigma(p_w, p_b - w_p, p_b)], \end{aligned} \quad (11)$$

where s_f , s_v , and s_p are safety sub-indicators for force, velocity, and energy flow $p_w = (e_f - f_r)(-\dot{e})$. User-defined safety bounds of velocity and energy flow are $\dot{e}_b$ and p_b , with w_f , w_v , and w_p the widths of variation. $\sigma(\alpha, \beta, \gamma)$ is a function that ensures a smooth impedance variation:

$$\sigma(\alpha, \beta, \gamma) = \begin{cases} 0, & \alpha < \beta \\ \frac{1}{2} + \frac{1}{2} \sin\left(\frac{\pi(\alpha-\gamma)}{\gamma-\beta} + \frac{\pi}{2}\right), & \alpha \in [\beta, \gamma] \\ 1, & \alpha > \gamma. \end{cases} \quad (12)$$

Here, s in (11) represents the current safety: $s = 0$ for danger, $s = 1$ for safety, $s \in [0, 1]$ for a situation in between.

Another property that further ensures interaction safety is passivity. To achieve the passivity, define the following storage function with the variable impedance stiffness $k(t)$ and the constant impedance mass m :

$$V = \frac{1}{2}k(t)e^2 + \frac{1}{2}m\dot{e}^2. \quad (13)$$

To ensure a passive system, a virtual energy tank V_t is designed to fully exchange the system energy as:

$$\begin{aligned} V_t|_{t=0} &= V_{t,0}, \quad V_t > 0 \quad \forall t, \\ \dot{V}_t &= d(t)\dot{e}^2 + f_r\dot{e} - \frac{1}{2}\dot{k}(t)e^2, \end{aligned} \quad (14)$$

where $V_{t,0}$ is the initial tank energy. When $V_t \rightarrow 0$, the elastic potential energy in V compensates for the further energy injection via a constrained safety indicator $\bar{s}$:

$$\begin{aligned} \bar{s}(0) &= s(0), \quad \bar{s}(t) \in [0, 1] \quad \forall t, \\ \dot{\bar{s}}(t) &= \begin{cases} \min(\dot{s}(t) + \gamma_s e_s(t), \\ \frac{2[d(t)\dot{e}^2 + f_r\dot{e}]}{ke^2}), & |e| > \epsilon \text{ and } V_t < \epsilon \\ \dot{s}(t) + \gamma_s e_s(t), & \text{otherwise,} \end{cases} \end{aligned} \quad (15)$$

where $e_s(t) = s(t) - \bar{s}(t)$, $\gamma_s > 0$ is a gain to drive $\bar{s}(t)$ towards $s(t)$ when the tank energy suffice, and $\epsilon = 10^{-4}$ avoids singularity when $|e| \rightarrow 0$. An extreme situation is

$V_t < \epsilon$, $\bar{s}(t) < \epsilon$, i.e., both tank energy and system potential energy are depleted. In this case, the interaction will be terminated by the safety stop to avoid any danger.

Finally, the actual variable impedance parameters are thus:

$$k(t) = \bar{s}k, \quad d(t) = \bar{s}(d - d_+) + d_+. \quad (16)$$

These variable parameters replace the constant parameters k and d in (7) for the actual MPC application.

The above developments of the control framework are now summarized in the form of a lemma.

Lemma 1. *With the linearized impedance tracking MPC (8) with variable impedance parameters $d(t)$, $k(t)$ from (16) based on $\bar{s}(t)$ from (15), the proposed control framework:*

- 1) *tracks an impedance model smoothly varying between the safe impedance model (10) and the desired impedance model (4).*
- 2) *remains passive with the virtual energy tank (14) and the passive safety indicator (15).*

Proof. Initially, s is tracked and constrained by $\bar{s}$ in (15). Thus, impedance parameters (16) vary between the desired impedance model (4) and the conservative impedance model (10) based on the interaction safety and passivity.

Regarding the passivity, differentiate V in (13) with (4):

$$\begin{aligned} \dot{V} &= m\dot{e}\ddot{e} + k(t)e\dot{e} + \frac{1}{2}\dot{k}(t)e^2 \\ &= -d(t)\dot{e}^2 - (f_r - f)\dot{e} + \frac{1}{2}\dot{k}(t)e^2. \end{aligned} \quad (17)$$

Denote V_f the external energy from f , and $\dot{V}_f = -f\dot{e}$. The augmented system energy is thus as $W = V + V_f + V_t$, whose energy flow rate $\dot{W}$ is computed with (14):

$$\begin{aligned} \dot{W} &= \dot{V} + \dot{V}_f + \dot{V}_t \\ &= -d(t)\dot{e}^2 - f_r\dot{e} + \frac{1}{2}\dot{k}(t)e^2 + \dot{V}_t = 0, \end{aligned} \quad (18)$$

indicating that the augmented system is passive.

Moreover, when $V_t < \epsilon$, from (15), one obtains:

$$\dot{V}_t = d(t)\dot{e}^2 + f_r\dot{e} - \frac{1}{2}\dot{s}(t)ke^2 \geq 0. \quad (19)$$

Here, the elastic potential energy in $\bar{s}(t)k$ takes the place of V_t for energy supply, maintaining $V_t \geq 0$ and the passivity not violated, until replenished or exhausted as well. ■

C. Predefined-Time Convergence

Despite the interaction safety, compliance, and passivity of the proposed control framework, there is no direct and explicit method to regulate the interaction convergence time. Such regulation is beneficial for numerous real-world applications. Thus, the proposed control framework is further improved for a practical PdT convergence. Previous works [20], [21] achieve PdT convergence by designing the system with a specific Lyapunov function derivative. Differently, in this work, the PdT convergence is achieved from the natural asymptotic convergence of IC via a time scaling $\nu(t)$:

$$\nu(t) = 1 + \left\lfloor g_\nu \frac{t - T_i}{T_p - t} \right\rfloor \in \mathbb{N}^+ \quad \forall t \in [T_i, T_p), \quad (20)$$

where T_i is the starting time, T_p is the predefined time of convergence, $\lfloor \cdot \rfloor$ is the floor function, $g_\nu > 0$ is the key meta-parameter to be tuned later in Subsection III-D. With the time scaling above, the original MPC-based control runs in the virtual time. The MPC prediction horizon is augmented as $h_p^* = h_p + \delta$, with $\delta = \nu(t) + n_c - 1$, and $n_c = 3$ as the order of controllability. The system propagation for δ intervals can be achieved within n_c intervals via a series of equivalent control input vector $\mathbf{u}_e \in \mathbb{R}^{n_c \times 1}$:

$$\begin{aligned} \mathbf{u}_e &= \bar{\mathbf{B}}_{n_c}^{-1}[(\mathbf{A}^\delta - \mathbf{A}^{n_c})\mathbf{x}(t) + \bar{\mathbf{B}}_\delta \bar{\mathbf{u}}_\delta]_{(n_c,1)}, \\ \bar{\mathbf{B}}_i &= [\mathbf{A}^{i-1}\mathbf{B} \quad \mathbf{A}^{i-2}\mathbf{B} \quad \cdots \quad \mathbf{B}], \\ \bar{\mathbf{u}}_i &= [u(t) \quad u(t+1) \quad \cdots \quad u(t+i-1)]^T. \end{aligned} \quad (21)$$

where $\lfloor \cdot \rfloor_{(r,c)} \in \mathbb{R}^{r \times c}$ is the truncated matrix of $\lfloor \cdot \rfloor$ with the beginning r rows and c columns remained. Elements in $\mathbf{u}_e$ are adopted as the actual control system input sequence, such that the same system propagation in δ intervals is now achieved in n_c time intervals.

Essentially, $\mathbf{u}_e$ maps from a virtual timeline with asymptotic convergence to the actual timeline with PdT convergence. This makes the desired impedance model (4) a ‘nominal’ impedance model, as the asymptotic convergence behavior only guides the virtual system in the virtual time.

Note that MPC constraints in (8) should be complied with in the actual timeline, not the virtual timeline. Thus, consider the mapping $\mu(t)$ from actual discrete timeline t to virtual discrete timeline t_v :

$$t_v = \mu(t) = t + g_\nu \left\lfloor \frac{t - T_i}{T_p - t} \right\rfloor T_s. \quad (22)$$

Given the small sampling interval, (22) is approximated in the continuous time with actual continuous timeline τ to virtual continuous timeline τ_v :

$$\tau_v = \mu_c(\tau) = \tau + g_\nu \frac{(\tau - T_i)}{(T_p - \tau)} T_s. \quad (23)$$

The continuous time derivative of $\mu_c(\tau)$ is:

$$\frac{d\mu_c(\tau)}{d\tau} = 1 + g_\nu \frac{(T_p - T_i)}{(T_p - \tau)^2} T_s = \frac{d\tau_v}{d\tau}. \quad (24)$$

Reform the MPC constraints together in the matrix form:

$$\left(\Psi_3 \frac{d^3}{d\tau^3} + \Psi_2 \frac{d^2}{d\tau^2} + \Psi_1 \frac{d}{d\tau} + \Psi_0 \right) e + \Psi_c \leq 0, \quad (25)$$

where $\Psi_3, \Psi_2, \Psi_1, \Psi_0, \Psi_c \in \mathbb{R}^{n_{ct} \times 1}$ with n_{ct} the number of constraints. From (24), reform (24) from τ to τ_v :

$$\begin{aligned} \left(\Psi_3 \left(\frac{d\tau_v}{d\tau} \right)^3 \frac{d^3}{d\tau_v^3} + \Psi_2 \left(\frac{d\tau_v}{d\tau} \right)^2 \frac{d^2}{d\tau_v^2} \right. \\ \left. + \Psi_1 \frac{d\tau_v}{d\tau} \frac{d}{d\tau_v} + \Psi_0 \right) e + \Psi_c \leq 0. \end{aligned} \quad (26)$$

These reformed constraints replace the original constraints in (8), such that the designed constraints are still complied with in the actual timeline regardless of the time scaling.

The safety while achieving PdT convergence is significant, especially with drastic disturbances. Such safety conditions can be indicated by $\bar{s}$ in (15). Thus, the following switching

mechanism is designed to ensure that the PdT convergence is targeted only when no danger exists, i.e., $\bar{s} = 1$:

- 1) Initialize with PdT convergence ON.
- 2) PdT convergence OFF if $\bar{s} < 1 - \epsilon$. The controller then follows that in **Lemma 1**.
- 3) PdT convergence ON again when $\bar{s} \leq 1 - \epsilon$ for T_{dw} intervals. Update $T_i \leftarrow t$ and $T_p \leftarrow t + (T_p - T_i + 1)$.
- 4) Go back to Step 2 until $t = T_p$.

Note that the minimal dwell time T_{dw} is the number of steps required for reactivation, which avoids the Zeno behavior for the switching mechanism.

With all the derivations so far, the EMSIC framework for a PdT convergence is proposed in the form of a theorem.

Theorem 1. By applying $\mathbf{u}_e$ in (21) obtained from $\bar{\mathbf{u}}_\delta$ in (8), with the MPC constraints being scaled according to (26), the actual system obeys the nominal impedance model at the beginning of the interaction task, and then achieves the convergence at the same steady state as controlled by the framework in **Lemma 1** at time t_{ss} with a predefined convergence time T_p such that $t_{ss} \leq T_p$.

Proof. When the control inputs computed from $\mathbf{u}_e$ are applied to the controlled system for n_c time steps, the system propagates exactly as the propagation of the nominal impedance model for δ time intervals. $\nu(t)$ in (20) has the property of $\nu(T_i) = 1$, and thus no time scaling since $\delta = n_c$. The controlled system follows the nominal impedance model.

Next, $\nu(t)$ has another property as follows:

$$\lim_{\frac{T_s}{T_p - T_i} \rightarrow 0} \sum_{t=T_i}^{T_p-1} \nu(t) = \int_{T_i}^{T_p} \nu(\tau) d\tau > \int_{T_i}^{T_p} g_\nu \frac{\tau - T_i}{T_p - \tau} d\tau = \infty. \quad (27)$$

This sum is the total number of time steps that t_v leads t . When $T_s/(T_p - T_i) \rightarrow 0$, t_v is in the lead of t by an infinity time steps, and the asymptotic convergence must have been achieved. This indicates the global PdT convergence can be achieved within the predefined time T_p . ■

D. Learning-Based Parameter Tuning

On one hand, as $T_s/(T_p - T_i) \rightarrow 0$ in (27) may not hold in practice, convergence errors may occur at T_p with an inappropriately conservative g_ν . On the other hand, an inappropriately aggressive g_ν may lead to dangerous interaction. For safety and PdT convergence in practical applications, g_ν in (20) must be carefully tuned, neither too small to achieve the PdT convergence nor too large for interaction safety.

Tuning g_ν fits into a contextual bandit problem. Here, the context vector $\chi \in \mathbb{R}^{n_{fc} + n_{fi}}$ includes (i) n_{fc} control features, e.g., predefined convergence time, and (ii) n_{fi} interaction features, e.g., environmental stiffness. The bandit arm $a \in \mathbb{R}$ is the selection of g_ν . The bandit reward (penalty) $r \in \mathbb{R}$ is designed as:

$$r(\chi, a) = -\frac{r_1}{\bar{e}_{s,1}} e_{pc} - \frac{r_2}{\bar{e}_{s,2}} f_{os} - \frac{r_3}{\bar{e}_{s,3}} \max |\dot{e}|. \quad (28)$$

Here, $e_{pc} = |e(T_p) - e(t_{end})|$ is the PdT convergence error with t_{end} the task ending time, $f_{os} = \max(\max |f| - f_r, 0)$

is the extra force beyond f_r , and $r_1, r_2, r_3 \in \mathbb{R}^+$ are weight coefficients. $\bar{e}_{s,1}, \bar{e}_{s,2}, \bar{e}_{s,3}$ are the mean values of $e_{pc}, f_{os}, \max|\dot{e}|$ from the simulation set. It should be noted that the reward remains negative in this work, and the tuning tries to enhance the reward towards zero.

In this work, a feedforward NN $\mathcal{N}$ is trained to map the contexts χ , and arms a to the predicted reward $\hat{r}$:

$$\hat{r} = \mathcal{N}(\chi, a). \quad (29)$$

Before the actual physical interaction task, via trial interactions with various physical environments, the trained NN is further fine-tuned to reduce the sim-to-real gap. During the physical fine-tuning, g_ν is selected by the following UCB algorithm to balance the exploration and exploitation:

$$g_\nu = \arg \max_a \text{UCB}(\chi, a, \mathcal{N}), \quad (30)$$

$$\text{UCB}(\chi, a, \mathcal{N}) = \mathcal{N}(\chi, a) + (g_0 + g_u) \sqrt{\frac{2 \log(t_{ft})}{n(a)}},$$

where t_{ft} is the time steps of fine-tuning, $n(a)$ is the number of this arm a being selected, g_0 is a fixed scaling coefficient that is selected based on the task uncertainty for robustness, and g_u is the varying scaling coefficient:

$$g_u = \sqrt{\frac{\nabla_{\theta_N} \mathcal{N}(\chi, a)^T \mathbf{S}_N^{-1} \nabla_{\theta_N} \mathcal{N}(\chi, a)}{w_N}}, \quad (31)$$

with the NN parameter vector θ_N , NN width w_N , and $\mathbf{S}_N$:

$$\mathbf{S}_N = \sum_{i=1}^{t_{ft}-1} \frac{\nabla_{\theta_N} \mathcal{N}(\chi, a) \nabla_{\theta_N} \mathcal{N}(\chi, a)^T}{w_N} \Big|_i. \quad (32)$$

The above UCB design assumes fixed parameters, e.g., NN depth, regularization factor, and step size, while a more generalized design can be found in previous works, e.g., [31]. It is also assumed that all possible contexts in physical interaction will be explored in simulation, and the same arm candidates are used for simulation and physical interaction. With the additional fine-tuning in the physical environment, the sim-to-real gap will be reduced.

The NN training and tuning are shown in **Algorithm 1**. Here, using simulation to train the NN before applying UCB in trial physical interactions reduces the number of required physical interactions N_i before obtaining the trained network $\mathcal{N}^*$, as physical trial interactions are more time-consuming compared to simulated interactions. Finally, $\mathcal{N}^*$ is frozen for faster responses and better real-time performance. The optimal meta-parameter g_ν^* can be obtained by $\mathcal{N}^*$:

$$g_\nu^* = a^* = \arg \max_a \mathcal{N}^*(\chi, a), \quad (33)$$

This tuning relies on the interaction environment feature in χ , which is initially unknown and may change as the interaction continues. Thus, tuning for g_ν^* based on the interaction environment in real time is necessary. Note that g_ν is a meta-parameter, and tuning it during the interaction will not directly affect the low-level system behavior, e.g., e or $\dot{e}$, and thus will not result in chattering or instability.

Algorithm 1 NN training and fine-tuning

- 1: **Input:** Context and arm sets χ_s, a_s , number of simulations N_s and trial physical interactions N_i
 - 2: **Initialization:** $\mathcal{N}(\chi, a) \rightarrow \hat{r}$ with random weights
 - 3: **for** $i_{sim} = 1, \dots, N_s$ **do**
 - 4: Randomly select $(\chi_{sim}, a_{sim}) \in \{\chi_s \times a_s\}$
 - 5: Simulate and record reward r_{sim} with (χ_{sim}, a_{sim})
 - 6: **end for**
 - 7: Train $\mathcal{N}$ with N_s data records of $(\chi_{sim}, a_{sim}, r_{sim})$
 - 8: **for** $i_{int} = 1, \dots, N_i$ **do**
 - 9: Move to a new location in the physical environment
 - 10: Observe χ , compute $a = \arg \max_a \text{UCB}(\chi, a, \mathcal{N})$
 - 11: apply a , observe r , and fine-tune $\mathcal{N}$ with (χ, a, r)
 - 12: **end for**
 - 13: **Output:** Trained NN $\mathcal{N}^*(\chi, a) \rightarrow \hat{r}$
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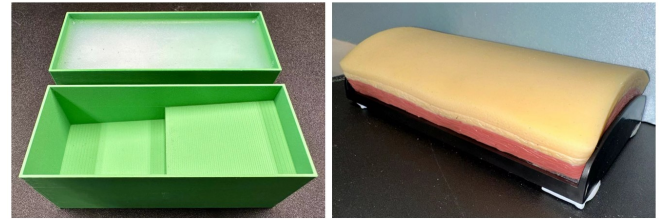


Fig. 3. Two silicone rubber models, A (left) and B (right). Model A is horizontally inhomogeneous due to different depths at different locations, and Model B is vertically inhomogeneous with different layers.

IV. EXPERIMENT

In this work, a 7-DoF KUKA iiwa robot is used for the experiments, together with a 2-finger Robotiq gripper to hold the interaction tool and an ATI force-torque sensor to measure the interaction force. The proposed learning-based control framework is realized in MATLAB, and the communication between MATLAB and the robot is achieved via ROS with a frequency of 50 Hz. As shown in Fig. 3, two complex silicone rubber models are adopted as the interaction environments in the experiments to fully simulate those complex real-world environments.

Some selections of parameters are listed below. Impedance parameters are $m = 1$, $d = 3$, $d_+ = 6$, $k = 0.2$, $z_r = 9$, and $f_r = 5$. The environment estimation threshold is $a_f = 0.5$ N. The MPC prediction and control horizons are $h_p = 5$, $h_c = 3$, and an MPC position constraint is set at 5 mm above the environment surface. For impedance variation, the safety bounds are $\dot{e}_b = 25$ mm/s and $p_b = 150$ mJ/s, and the widths of variation are $w_f = 1$ N, $w_v = 5$ mm/s, and $w_p = 10$ mJ/s. The energy tank is initialized with $V_{t,0} = 100$ mJ. The safety indicator tracking gain is $\gamma_s = 2$. The predefined convergence time is $T_p = 200$, i.e., 4 seconds. The memory window for the switching mechanism is $T_{dw} = 5$.

A. Implementation of the Tuning Schedule

The NN is initially prepared for tuning the key meta-parameter g_ν . Specifically, here, the NN takes bandit context (environment stiffness k_{env} and predefined time T_p) and arm (g_ν) as inputs, and outputs the predicted reward. The structure includes: an input layer that takes the 3 inputs

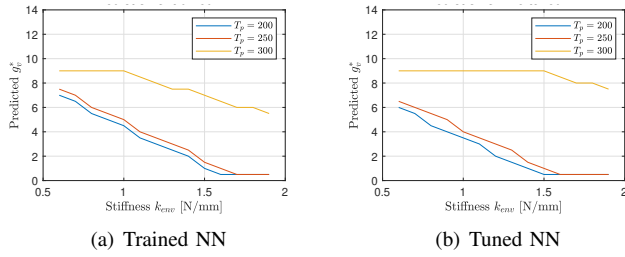


Fig. 4. Predictions on the optimal arm g_v^* with different contexts.

with normalization, a hidden layer with 15 neurons with normalization, a ReLU layer, and an output layer.

Following **Algorithm 1**, data from 10,000 simulation runs based on our proposed framework form up the simulation dataset. The aforementioned three disturbance scenarios have equal probabilities of occurring in the simulation runs, with randomized occurrence times and disturbance distances. Here, bandit contexts and arm are also randomized in the range of $k_{env} \in [0.5, 2.5]$, $T_p \in [150, 350]$, and $g_v \in [0.5, 9]$, and the reward follows (28) with $r_1 = 1$, $r_2 = r_3 = 0.3$, $\bar{e}_{s,1} = 0.280$, $\bar{e}_{s,2} = 0.494$, $\bar{e}_{s,3} = 6.232$. The NN is then trained via stochastic gradient descent algorithm for 20 epochs with learning rate 0.01 and batch size 100.

The trained NN is then fine-tuned in the physical environment. Model A is fully used here, with the interaction location randomly selected (thus physical environment stiffness also randomized within the range of $k_{env} \in [0.62, 2.41]$). Also, randomize $T_p \in [150, 350]$, and the arm is selected via (30). The dataset contains data from 240 physical interaction runs. To prevent overfitting, fine-tuning contains 2 epochs only, with a learning rate of 0.001 and batch size of 60.

Prediction results by trained and fine-tuned NNs on g_v^* with different contexts k_{env} and T_p are shown in Fig. 4. According to the figure, the predicted optimal arm becomes more aggressive as T_p increases and k_{env} decreases. This matches with the common sense: (i) a softer environment requires a larger moving distance until the convergence, and thus a more aggressive meta-parameter, and (ii) a longer T_p prefers a more aggressive meta-parameter as the mapping of timeline will be less drastic. The fine-tuning process also slightly modifies the NN to reduce the sim-to-real gap, which is possibly due to the limited knowledge and estimation of real-world environments and sensor noises.

B. Experimental Results

To showcase the performance of our proposed framework and the learning-based tuning, the following six control approaches are compared by physical experiments: (i) IC [3], (ii) MPIC [15], (iii) our proposed framework without PdT convergence, and our proposed framework with PdT convergence with the key meta-parameter g_v : (iv) not tuned (where $g_v = 5$ by default), (v) manually tuned once, and (vi) tuned with NN. Here, IC and MPIC follow the desired impedance model (4). To further demonstrate the generalizability of our framework, experiments are designed with different disturbance scenarios: (i) no disturbance, (ii)

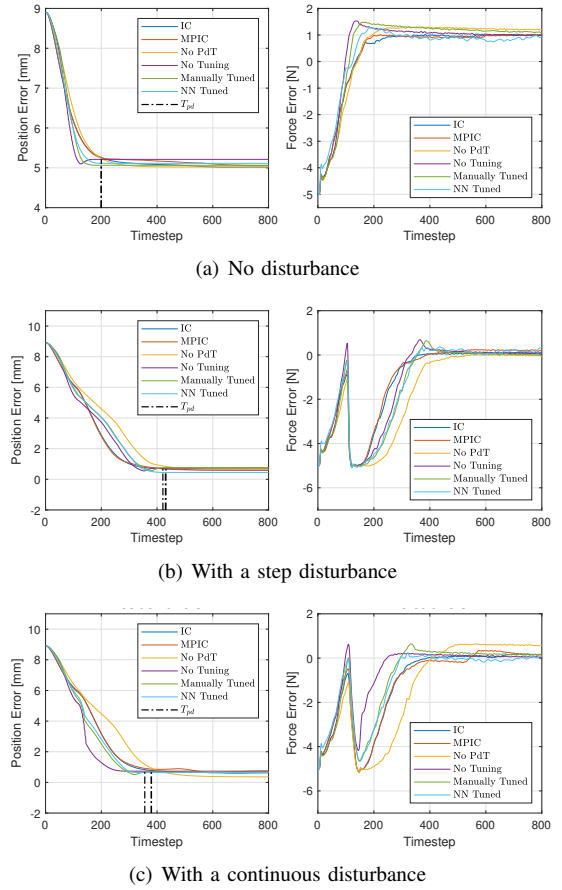


Fig. 5. Experimental results on Model A.

step disturbance, and (iii) continuous disturbance. The above experiments are conducted on both Models A and B.

The experimental results are shown in Fig. 5 and 6. The results are compared by the rewards r from (28) based on the convergence error e_{pc} , the extra force f_{os} , and the maximum velocity $\max |\dot{e}|$. They are all listed in Table I.

From the results, the proposed PdT control successfully achieves the PdT convergence in all experiments by driving the robot to the final steady state with a convergence error of less than 0.1 mm, which IC, MPIC, and our framework without PdT convergence all fail to achieve. Also, the NN-based tuning adjusts g_v to become appropriately more aggressive (a larger $\max |\dot{e}|$) or conservative (a larger e_{pc}). In all experiments, the highest reward is obtained by the framework with NN-based tuning of g_v compared to those without tuning or with manual tuning only, and the improvement of NN-based tuning is significant, especially in scenarios with disturbances. This validates that the proposed framework with NN-based tuning can safely achieve PdT convergence in disturbance-rich interaction tasks.

C. Case Study: Ultrasound Scanning

To demonstrate the functionality of our proposed framework for real-world applications, the PdT control framework is applied to ultrasound scanning on a human arm. The arm would move up and down randomly during the interaction, and such motions can be considered disturbances. The con-

TABLE I
EXPERIMENTAL RESULT DATA

Method	Disturbance	Model A				Model B			
		e_{pc}	f_{os}	$\max \dot{e} $	$r \uparrow$	e_{pc}	f_{os}	$\max \dot{e} $	$r \uparrow$
IC	No	0.243	1.030	1.755	-1.578	0.442	0.318	1.964	-1.866
MPIC	No	0.193	1.031	1.765	-1.401	0.422	0.381	2.071	-1.837
Ours (No PdT)	No	0.214	1.309	1.601	-1.636	1.218	0.113	1.599	-4.494
Ours+PdT (No Tuning)	No	0.000	1.529	5.467	-1.192	0.000	0.983	8.520	-1.008
Ours+PdT (Manual Tuning)	No	0.000	1.482	4.246	-1.105	0.000	0.769	10.854	-0.990
Ours+PdT (NN Tuning)	No	0.000	1.279	3.383	-0.940	0.030	0.650	7.095	-0.844
IC	Step	0.139	0.138	1.935	-0.672	0.243	0	1.969	-0.963
MPIC	Step	0.126	0.247	1.981	-0.696	0.167	0	2.083	-0.695
Ours (No PdT)	Step	0.213	0.024	1.601	-0.852	0.445	0	1.600	-1.665
Ours+PdT (No Tuning)	Step	0.000	0.690	5.443	-0.681	0.073	0.001	4.609	-0.484
Ours+PdT (Manual Tuning)	Step	0.000	0.647	4.254	-0.598	0.000	0	9.615	-0.463
Ours+PdT (NN Tuning)	Step	0.000	0.360	3.591	-0.392	0.022	0	6.147	-0.376
IC	Continuous	0.221	0.187	1.913	-0.995	0.235	0	1.979	-0.933
MPIC	Continuous	0.183	0.346	1.890	-0.954	0.334	0	2.077	-1.292
Ours (No PdT)	Continuous	0.343	0.647	1.603	-1.675	0.496	0	1.431	-1.838
Ours+PdT (No Tuning)	Continuous	0.000	0.628	21.057	-1.395	0.000	0	21.768	-1.048
Ours+PdT (Manual Tuning)	Continuous	0.000	0.641	11.565	-0.947	0.022	0	9.230	-0.520
Ours+PdT (NN Tuning)	Continuous	0.000	0.238	10.365	-0.643	0.026	0	7.693	-0.462

TABLE II
CASE STUDY RESULT DATA

	e_{pc}	f_{os}	$\max \dot{e} $	$r \uparrow$
1st Manual Tuning	0.000	17.612	11.448	-11.254
2nd Manual Tuning	0.009	17.358	4.096	-10.778
3rd Manual Tuning	0.000	15.638	6.642	-9.824
NN Tuning	0.000	15.190	6.758	-9.557

trol framework is applied along z direction with the same settings as in previous experiments, and the robot moves with a constant speed along $-x$ direction. Initially, the proposed framework is tested with manual tuning of g_ν by an expert for 3 times in a row to improve the final reward, where the expert tunes g_ν after analyzing the previous experiment. Then, the framework is tested once with NN-based tuning with a similar disturbance, where the NN tunes g_ν right after estimating the context in real-time.

The results are in Fig. 7, and the reward r , together with e_{pc} , f_{os} , and $\max |\dot{e}|$, are in Table II. It can be observed that although the reward has been gradually improved by the manual tuning, the reward from the NN-based tuning is still greater than that from the manual tuning after three trials by an expert. Thus, it can be concluded that the NN-based tuning successfully selects the appropriate meta-parameter g_ν , and the proposed PdT control framework safely achieves the PdT convergence by enhancing the convergence accuracy while reducing the interaction force and velocity.

The interaction experiments in this case study with manual tuning and with NN-based tuning can also be viewed in the video file submitted together with this manuscript. From the video, it can be observed that the time consumption of three manual tuning is much greater than one NN-based tuning.

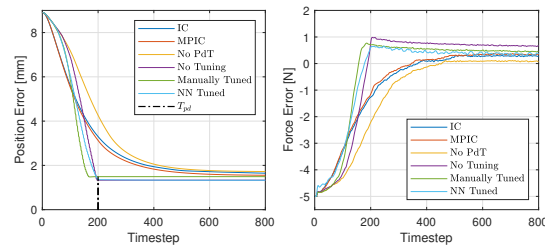
V. CONCLUSION

In this paper, a novel learning-based model predictive impedance control framework is introduced for achieving safe interactions and converging within a predefined time. Initially, the impedance controller is realized with MPC for constraint compliance and with impedance variation for safety and passivity guarantee. The controller is then extended with a time scaling function to achieve the PdT convergence. An NN is trained to predict the optimal key meta-parameter for predefined time convergence based on the RL reward from convergence error, interaction force overshoot, and maximum end-effector velocity.

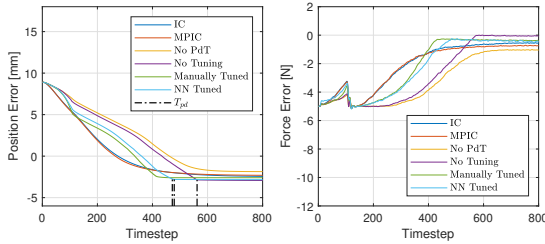
Experiments compare the performance of the proposed learning-based control framework with other frameworks without PdT convergence, or without learning-based tuning. Experimental results show that the proposed framework successfully achieves PdT convergence during interaction tasks while maintaining safety. Learning-based meta-parameter tuning also successfully increases the designed reward compared to other frameworks without tuning or with manual tuning by an expert.

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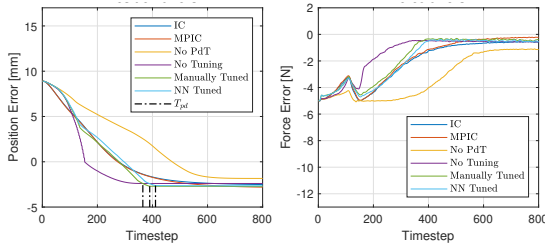
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(a) No disturbance



(b) With a step disturbance



(c) With a continuous disturbance

Fig. 6. Experimental results on Model B.

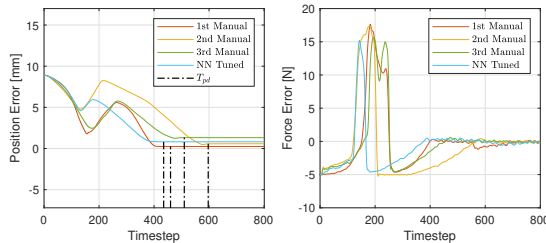


Fig. 7. Results of case study on a real human arm.

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